# Geometry and Vectors

1. Find the distance between $(3, 1)$ and $(8, -11)$.

2. Find the midpoint of the segment joining $(7, -5)$ and $(5, 0)$.

3. Find the slope of the line through $(7, -5)$ and $(12, -1)$.

4. Find an equation of the line through $(-3, 3)$ and $(-4, -3)$.

5. Find the distance from $(3, -4)$ to the line $3x - 4y = 0$.

6. A line has slope $-3$. Find the slope of a perpendicular line.

7. Find the area of the triangle with vertices $(1, 1)$, $(9, 1)$, and $(8, 3)$.

8. Find the centroid of the triangle with vertices $(-5, 1)$, $(4, 3)$, and $(6, 8)$.

9. A right triangle has known side lengths 7 and 24. Find the missing hypotenuse.

10. Write the equation of the circle with center $(2, 2)$ and radius $8$.

11. A circle has radius $4$ and central angle $180^\circ$. Find the arc length and sector area.

12. For a regular 6-gon, find the sum of its interior angles and each interior and exterior angle.

13. A cylinder has radius 6 and height 3. Find its exact volume.

14. Find the image of $(6, -3)$ after reflection across the x-axis.

15. Find the vector from $(5, 3)$ to $(-1, 0)$.

16. Classify the vectors as parallel, perpendicular, or neither: $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

17. Find the angle between $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 3 \end{pmatrix}$.

18. Compute $\begin{pmatrix} 5 \\ -4 \\ -5 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ -5 \end{pmatrix}$.

19. Write a vector equation for the line through $(-1, -1)$ with direction $\begin{pmatrix} -4 \\ -1 \end{pmatrix}$.

20. Find the plane through $(4, 5, -3)$ with normal vector $\begin{pmatrix} -3 \\ -1 \\ 1 \end{pmatrix}$.

# Solutions

1. $d = \sqrt{169} = 13$.
2. The midpoint is $\left(6,\ -\frac{5}{2}\right)$.
3. $m = \frac{4}{5} = \frac{4}{5}$.
4. In standard form, $6x - y + 21 = 0$.
5. $d = \frac{25}{\sqrt{25}} = 5$.
6. Perpendicular slopes are negative reciprocals: $\frac{1}{3}$.
7. $A = \frac{1}{2}|8 \cdot 2| = 8$.
8. Average the coordinates: $\left(\frac{5}{3}, 4\right)$.
9. $c = \sqrt{7^2 + 24^2} = 25$.
10. $(x-2)^2 + (y-2)^2 = 64$.
11. Arc length $4\pi$; sector area $8\pi$.
12. Sum: $720^\circ$; each interior angle: $120^\circ$; each exterior angle: $60^\circ$.
13. Volume $108\pi$.
14. The image is $(6, 3)$.
15. Subtract start from end: $\begin{pmatrix} -6 \\ -3 \end{pmatrix}$.
16. They are perpendicular. $u \cdot v = 0$.
17. $\cos\theta = \frac{u \cdot v}{\|u\|\|v\|}$, so $\theta = 135^\circ$.
18. The cross product is $\begin{pmatrix} 15 \\ 15 \\ 3 \end{pmatrix}$.
19. $r(t) = \begin{pmatrix} -1 \\ -1 \end{pmatrix} + t\begin{pmatrix} -4 \\ -1 \end{pmatrix}$.
20. An equation is $-3x - y + z + 20 = 0$.